

Search Arbitrage Platform — Product Document

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1. Executive Summary

A self-hosted search arbitrage platform that enables media buyers to launch Facebook ads, automatically generates monetized article pages, handles AdSense AFS revenue attribution at the individual ad level, and provides real-time ROI dashboards.

Core Value Proposition: Media buyers get a streamlined, search-arb-specific Facebook ad launcher with automatic article generation, revenue tracking, and optimization tools — all in one platform.

Domain Architecture:

- **Platform Domain** — Dashboard, ad launcher, admin panel (e.g., `app.yourdomain.com`)
- **Article Domain** — Hosted articles with AFS widgets (must be AdSense-approved, e.g., `articles.yourdomain.com`)
- **Redirect Domain** — Separate domain for redirect engine / cloaking (e.g., `go.anotherdomain.com`)

2. Business Model

```
[Media Buyer] → Launches Facebook Ad → Spends $X
               ↓
[User clicks ad] → Redirect Engine → Article Page (with AFS widget)
               ↓
[User clicks AFS search ad] → Revenue $Y attributed via Channel
               ↓
[Dashboard] → Shows Spend $X vs Revenue $Y → ROI = (Y-X)/X
```

Revenue Sources:

- AdSense AFS revenue from search ad clicks on article pages
- Platform takes a configurable revenue cut per media buyer (admin sets % per user, e.g., 30% means buyer sees 70% of actual revenue)
- Revenue shown to buyer = Actual AdSense Revenue x (1 - revenue_cut_pct)
- Admin dashboard shows actual revenue + buyer's visible revenue + platform margin

Key Metrics:

- ~50 media buyers
- ~100s of ads launched daily
- ~1,500 active ads at any time
- 2,000 AdSense AFS channels for attribution
- Revenue pulled hourly for near real-time ROI

3. User Roles & Permissions

3.1 Media Buyer (User)

PermissionAccess
|-----|-----|
Sign up / login
Connect Facebook account
View own ad accounts, pages, pixels
Create campaigns, adsets, ads
Upload creatives (image/video)
View own dashboard (spend, revenue, ROI)
Pause/resume own ads
Set redirect URL for non-ad traffic
View other buyers' data
Manage channels
Approve ads

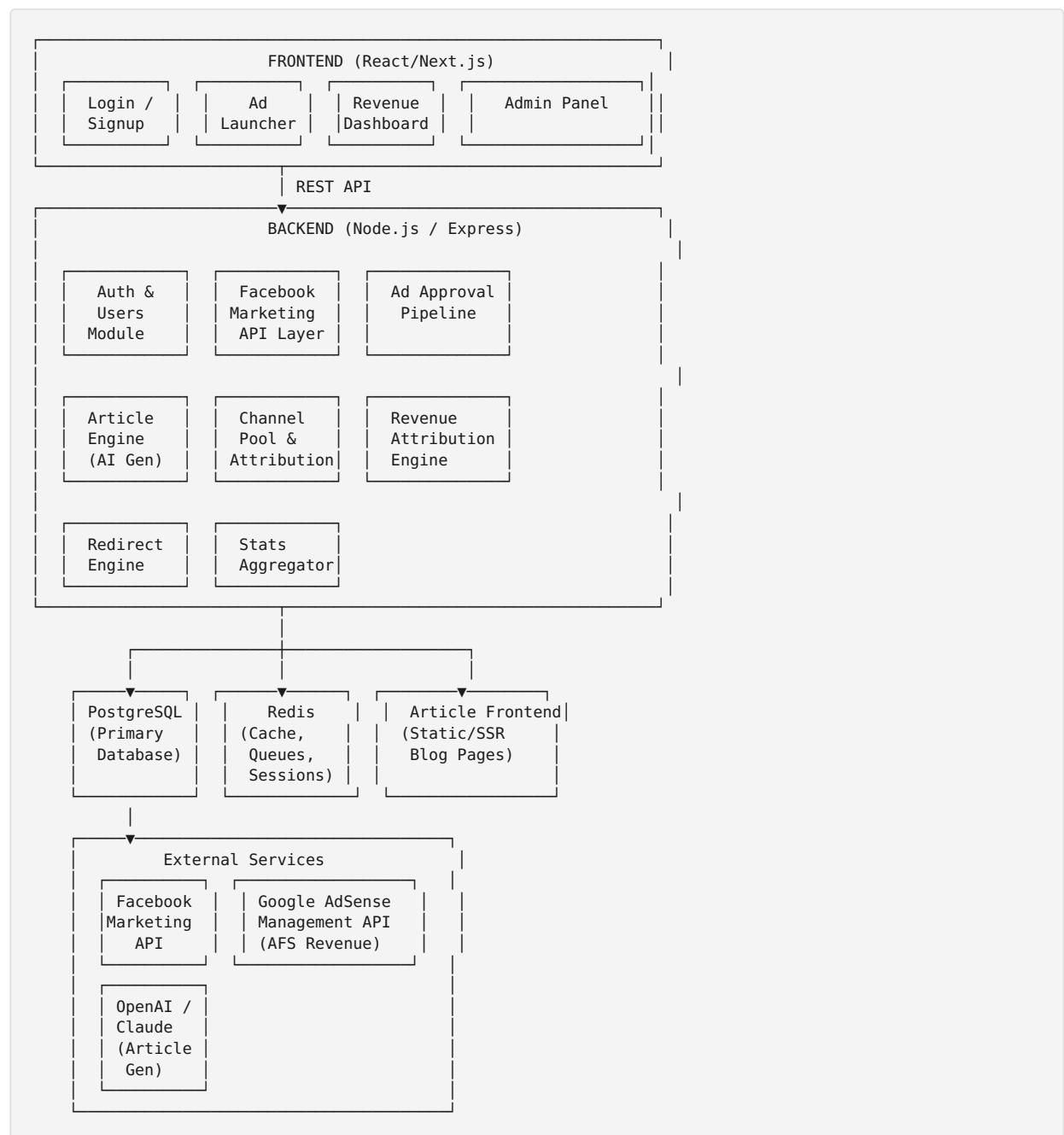
3.2 Admin

PermissionAccess
|-----|-----|
Approve/reject user signups
Approve/reject/auto-approve ads

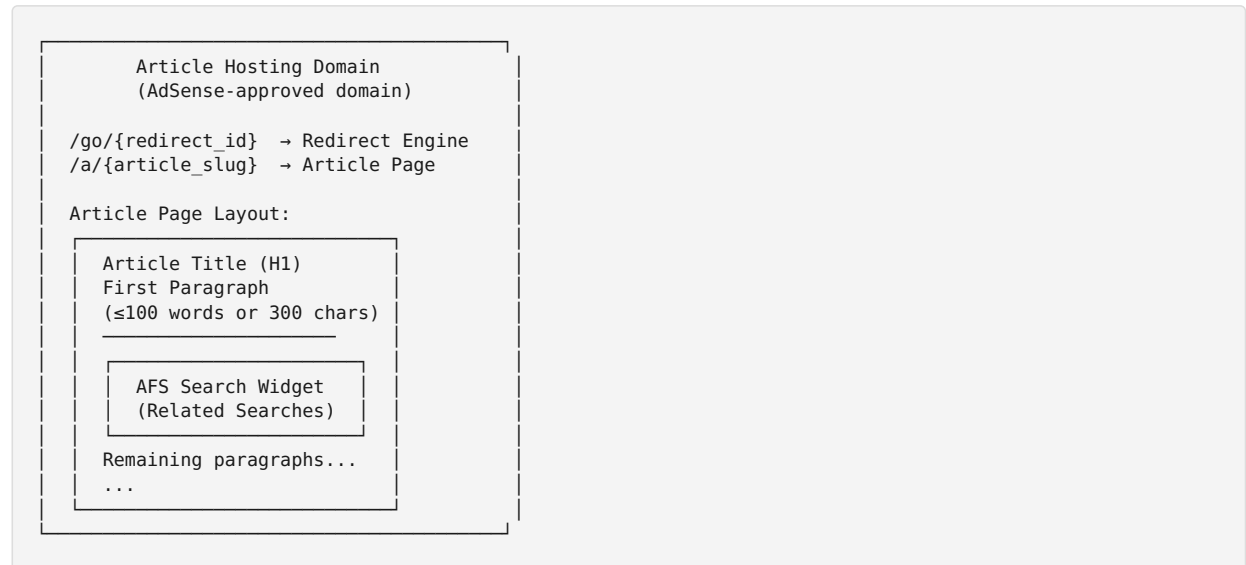
Override any buyer's ad (pause/kill)
Manage channel pool (add/remove channels)
View all users' stats + platform totals
Per-user revenue breakdown
Manage domains, global settings
Manage redirect domains + assign to buyers
Manage article templates
View/edit any article
Set revenue cut % per media buyer

4. System Architecture

4.1 High-Level Architecture



4.2 Article Frontend Architecture



5. Module Specifications

5.1 User Management

5.1.1 Signup Flow

```
User fills signup form
→ Name, Email, Password
→ Account created with status: PENDING
→ Admin gets notification
→ Admin approves/rejects
→ User gets email notification
→ If approved: can log in and use platform
→ If rejected: sees "Account not approved" message
```

5.1.2 Authentication

- JWT-based authentication
- Access token (15 min) + Refresh token (7 days)
- Password hashing: bcrypt (12 rounds)

- Optional 2FA (future scope)

5.1.3 User States

- **PENDING** — just signed up, awaiting admin approval
- **ACTIVE** — approved, can use platform
- **SUSPENDED** — admin suspended account
- **REJECTED** — admin rejected signup

5.2 Facebook Integration

5.2.1 Facebook App Setup

- Create Facebook App at developers.facebook.com

- App Type: Business

- Required Permissions:

- `ads_management` — create/manage ads
- `ads_read` — read ad performance
- `pages_show_list` — list pages
- `pages_read_engagement` — page info
- `business_management` — ad account access

- App Review required for production access

5.2.2 OAuth Flow

```
User clicks "Connect Facebook"
  → Redirect to Facebook OAuth dialog
  → Permissions: ads_management, ads_read, pages_show_list, business_management
  → User grants access
  → Facebook redirects back with auth code
  → Backend exchanges code for long-lived token (60 days)
  → Store encrypted token in DB
  → Fetch user's ad accounts, pages, pixels
  → Store in DB linked to user
  → Token refresh cron runs every 50 days
```

5.2.3 Data Synced from Facebook

- **Ad Accounts:** ID, name, currency, timezone, status
- **Pages:** ID, name, associated Instagram account
- **Pixels:** ID, name, configured events (lander, search, adclick)
- **Campaign performance:** impressions, clicks, spend (pulled hourly)

5.2.4 Token Management

- Long-lived tokens stored encrypted (AES-256)
- Auto-refresh before expiry (cron: every 50 days)
- If token becomes invalid → notify user to reconnect
- Support multiple ad accounts per user

5.3 Ad Launcher

5.3.1 Campaign Creation Flow

The ad launcher follows Facebook's Campaign → Adset → Ad hierarchy.

Step 1: Campaign

FieldTypeRequiredNotes

|-----|-----|-----|-----|

Campaign NametextAuto-generated if empty

Ad AccountselectFrom connected accounts

ObjectivesselectDefault: OUTCOME_ENGAGEMENT
 Special Ad Categoriesmulti-selectEMPLOYMENT, HOUSING, CREDIT, etc.
 Campaign BudgetnumberEither campaign OR adset budget
 Budget TypeselectDaily / Lifetime
 Bid StrategysselectLOWEST_COST_WITHOUT_CAP (default), COST_CAP, BID_CAP
Keywordstext/tagsFor article generation + AFS (campaign-level)
RAC ValuetextreferredAdCreative (campaign-level)
Fallback URLtextNon-ad traffic destination + display link
Traffic SplitconfigMulti-website split for ad traffic (see 5.7.4)

Step 2: Adset

FieldTypeRequiredNotes
 |-----|-----|-----|-----|
 Adset NametextAuto-generated if empty
 Adset BudgetnumberIf no campaign budget
 Optimization GoalselectDefault: OFFSITE_CONVERSIONS
 Billing EventsselectDefault: IMPRESSIONS
 PixelselectFrom ad account's pixels
 Conversion EventsselectFrom pixel's configured events
 Countriesmulti-selectGEO targeting
 Excluded Countriesmulti-select
 Age MinnumberDefault: 18
 Age MaxnumberDefault: 65
 GenderselectAll / Male / Female
 Placementsmulti-selectDefault: Automatic
 Device Platformsmulti-selectDesktop / Mobile / All
 Mobile OSmulti-selectiOS / Android / All
 Publisher Platformsmulti-selectFacebook, Instagram, Messenger, AN
 Start TimedatetimeDefault: immediately
 Advantage+toggleAdvantage campaign budget
 Attribution Windowselect1-day click, 7-day click, etc.

Step 3: Ad(s) — Multiple ads can be added per adset

FieldTypeRequiredNotes
 |-----|-----|-----|-----|
 Ad NametextAuto-generated if empty
 Facebook PagesselectFrom ad account's pages
 Instagram AccountselectIf Instagram placement selected
 Creative (Image/Video)file uploadjpg, png, webp, mp4
 HeadlinetextMax 40 chars
 Primary Texttextareamax 125 chars recommended
 Descriptiontext
 Display LinkautolockedAuto-populated from fallback URL (cloak URL where non-ad traffic goes)
 Call to ActionselectLEARN_MORE (default), SHOP_NOW, etc.
 Destination URLautoAuto-generated: redirect URL
 ~~~Keywords~~~—Moved to campaign level  
 ~~~RAC Value~~~—Moved to campaign level  
Fallback URLtextnoWhere non-ad traffic goes (overrides account-level default)
BeneficiarytextnoEU ad transparency

5.3.2 Prefilled / Hidden Defaults (Search Arb Specific)

These are locked or prefilled to streamline the search arb workflow:

SettingDefault ValueVisibleEditable
 |-----|-----|-----|-----|
 ObjectiveOUTCOME_SALES
 OptimizationOFFSITE_CONVERSIONS
 Destination URLAuto-generated redirect URL
 Link PreviewAuto from article

5.3.3 Multi-Ad Support

Within one adset, the buyer can add multiple ads:

- Each ad has its own creative, headline, primary text, keywords, RAC
- Each ad gets its own article (or matched to existing)
- Each ad gets its own channel assignment
- Each ad gets its own redirect URL

5.3.4 Submission → Approval Pipeline

```

Buyer submits ad(s)
  → Status: PENDING_APPROVAL
  → Admin notified
  → Admin reviews (or auto-approve if enabled)
  → If APPROVED:
    → Article Engine generates/matches article
    → Channel Pool assigns channel
    → Redirect URL generated
    → Facebook Marketing API creates Campaign → Adset → Ad(s)
    → Status: LAUNCHING → ACTIVE
  → If REJECTED:
    → Buyer notified with reason
    → Status: REJECTED

```

5.4 Ad Approval System

5.4.1 Approval Modes

ModeDescriptionConfig

|-----|-----|-----|

Manual ReviewAdmin reviews each adDefault

Auto ApproveAll ads auto-approved immediatelyToggle in admin settings

AI ReviewAI checks content compliance (future)Not implemented in v1

5.4.2 Approval States

```

PENDING_APPROVAL → APPROVED → PROCESSING → LAUNCHING → ACTIVE
                  → REJECTED

APPROVED → BATCHED (Facebook API rate limit queue)
APPROVED → QUEUED_NO_CHANNEL (waiting for free channel)

ACTIVE → PAUSED (by buyer or admin)
ACTIVE → STOPPED (no impressions 24h – channel released)
ACTIVE → META_REJECTED (Facebook rejected post-launch)
PAUSED → ACTIVE (resumed – real-time channel check)
PAUSED → QUEUED_NO_CHANNEL (no channel available on resume → auto-paused + buyer notified)

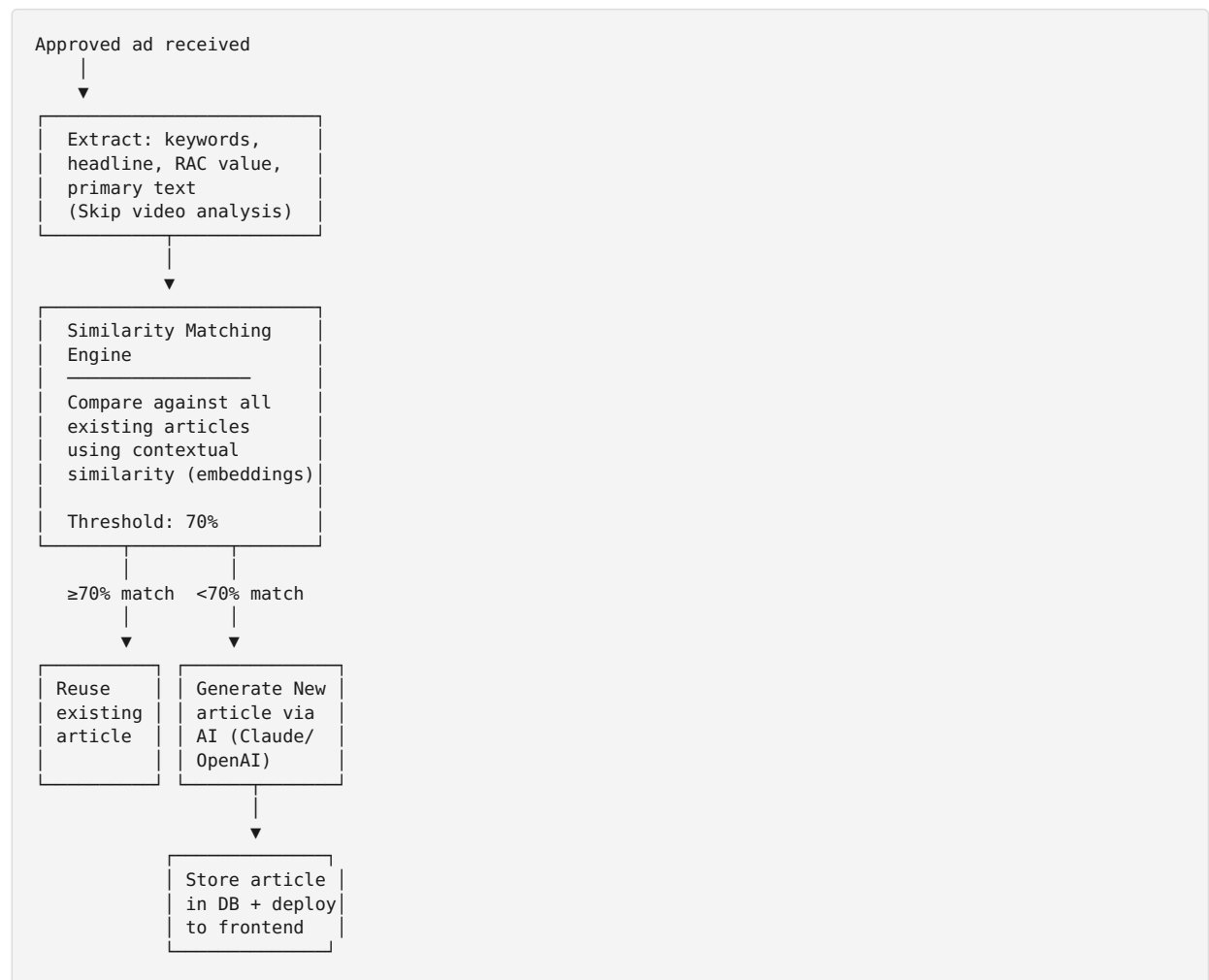
```

5.4.3 Meta Rejection Detection

- Cron job runs every 30 minutes
- For each ACTIVE ad, check Facebook API for `effective_status`
- If status = `DISAPPROVED` or `WITH_ISSUES` :
 - Update internal status to `META_REJECTED`
 - Store rejection reason from API
 - Notify buyer
 - Release channel (after midnight)

5.5 Article Engine

5.5.1 Article Generation Flow



5.5.2 Similarity Matching Engine

- **Input:** keywords + headline + RAC value from new ad
- **Method:** Text embeddings (OpenAI `text-embedding-3-small` or similar)
- **Storage:** Vector stored per article in DB
- **Comparison:** Cosine similarity between new ad's embedding and all existing articles
- **Threshold:** ≥ 0.70 = match → reuse article
- **On match:** Link the ad to the existing article (no new generation)

5.5.3 Article Generation Spec

AI Prompt Template:

Generate a blog article based on:

- Keywords: {keywords}
- Headline/Topic: {headline}
- Primary Text: {primary_text}

- RAC Value: {rac_value}

Requirements:

- 800-1200 words
- Text only, no images
- First paragraph must be ≤100 words OR ≤300 characters (whichever is shorter)
- Informative, engaging, SEO-friendly
- Natural tone, not salesy
- Split into clear paragraphs with subheadings
- The article should be contextually related to the keywords

Compliance Pass:

After initial generation, the article is run through a compliance engine:

1. Admin sets compliance instructions in Admin Panel > Compliance Settings
2. Stored in `platform_settings.compliance_prompt`
3. Article + compliance prompt sent to AI for rewrite
4. Both raw and compliant versions stored for audit
5. Compliant version is published

Example compliance prompt: "Remove health claims. Don't promise specific outcomes. Add disclaimers where needed."

Output Structure:

```
{
  "title": "Article Title",
  "slug": "article-title-slug",
  "firstParagraph": "First 100 words / 300 chars...",
  "remainingContent": "Rest of the article...",
  "metaDescription": "SEO meta description",
  "keywords": ["kw1", "kw2"],
  "embedding": [0.123, 0.456, ...],
  "wordCount": 950
}
```

5.5.4 Article Page Layout

{title}

{firstParagraph}

```
{remainingContent}
```

5.5.5 AFS Widget Parameters (Dynamic)

These are injected via the redirect engine as URL parameters:

ParameterSource

|-----|-----|

channel From channel pool (assigned to this ad)

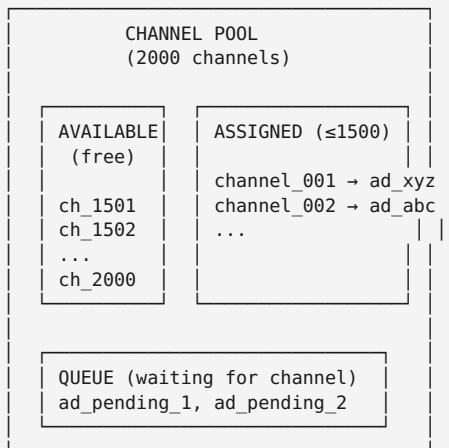
query / keywords From ad's keyword list

rac From ad's RAC value

adtest Toggle for testing mode

5.6 Channel Pool & Attribution System

5.6.1 Channel Pool Management



5.6.2 Channel Assignment Rules

RuleDescription

|-----|-----|

AssignmentWhen campaign is APPROVED and ready to launch, assign an available channel to the CAMPAIGN

Lock PeriodChannel stays with ad for entire calendar day (IST)

ReleaseChannel released when: campaign has 0 impressions for 24h AND next calendar day (IST) has started

Pause/Resumelf campaign paused and resumed on SAME day → keeps same channel

Pause/Resumelf ad paused, and resumed on DIFFERENT day → may get different channel (if original was released)

Queuelf no channels available → ad enters queue, launches when channel frees up

Queue PriorityFIFO — first queued, first assigned

Resume LogicOn campaign resume: check if channel still assigned. If released, assign new one immediately (real-time). If no channel free, auto-pause campaign + notify buyer + queue

Auto-Pauself campaign resumes but no channel available: pause on Facebook, set QUEUED_NO_CHANNEL, auto-resume when channel frees up

Midnight RolloverAt midnight IST: check all assigned channels, release those with inactive ads

5.6.3 Channel-Revenue Attribution Flow

Every hour (cron):





5.6.4 Historical Attribution

Since channels are reused across days, we maintain a history:

```

channel_assignment_history:
channel_id  ad_id  assigned_date  released_date
|-----|-----|-----|-----|
ch_001ad_12026-05-262026-05-28
ch_001ad_52026-05-29NULL (active)
  
```

Revenue for `ch_001` on `2026-05-26` → attributed to `ad_1`

Revenue for `ch_001` on `2026-05-29` → attributed to `ad_5`

5.6.5 Midnight Cleanup Cron (IST)

Runs daily at 00:05 IST:

```

For each assigned channel:
  If ad.status IN (STOPPED, META_REJECTED, PAUSED for >24h):
    Release channel back to pool
    Update channel_assignment_history.released_date

For each queued ad:
  If available channels > 0:
    Assign channel
    Launch ad on Facebook
    Update status to ACTIVE
  
```

5.7 Redirect Engine

5.7.1 Redirect URL Format

```
https://{redirect-domain}/go/{redirect_id}
```

Where `redirect_id` is a unique short ID mapped to an ad.

5.7.2 Redirect Logic



5.7.3 URL Parameter Mapping

ParameterSourcePurpose

| Parameter | Source | Purpose |
|-----------|-------------------------------|--|
| ch | Channel from poolAdSense | AFS channel attribution |
| q | Ad's keywords | AFS search query |
| rac | Ad's RAC value | referredAdCreative for AFS |
| ref | Redirect ID | Internal tracking |
| styleId | User's custom search style ID | AFS widget styling |
| px | Meta Pixel ID | Dynamic pixel firing |
| pxe | Pixel conversion event | Default: "search" — fires on AFS click |

5.7.4 Offer Testing / Traffic Split

Campaign-level traffic split configuration:

SettingDescription

| Setting | Description |
|--------------------|--|
| Non-Ad Traffic URL | Single fallback URL (where organic/bot traffic goes) |
| Ad Traffic Split | Multiple websites with weight percentages (must sum to 100%) |

Example: Website 1 at 40%, Website 2 at 60%

Redirect engine for ad traffic:

1. Weighted random selection of destination website
2. Append all params (channel, keywords, RAC, styleId, pixel) to selected URL
3. 302 redirect to selected destination

New table: `campaign_traffic_splits`

- campaign_id, website_url, weight_pct, afs_config (json), status

- Weights must sum to 100%

5.7.5 Fallback URL Configuration

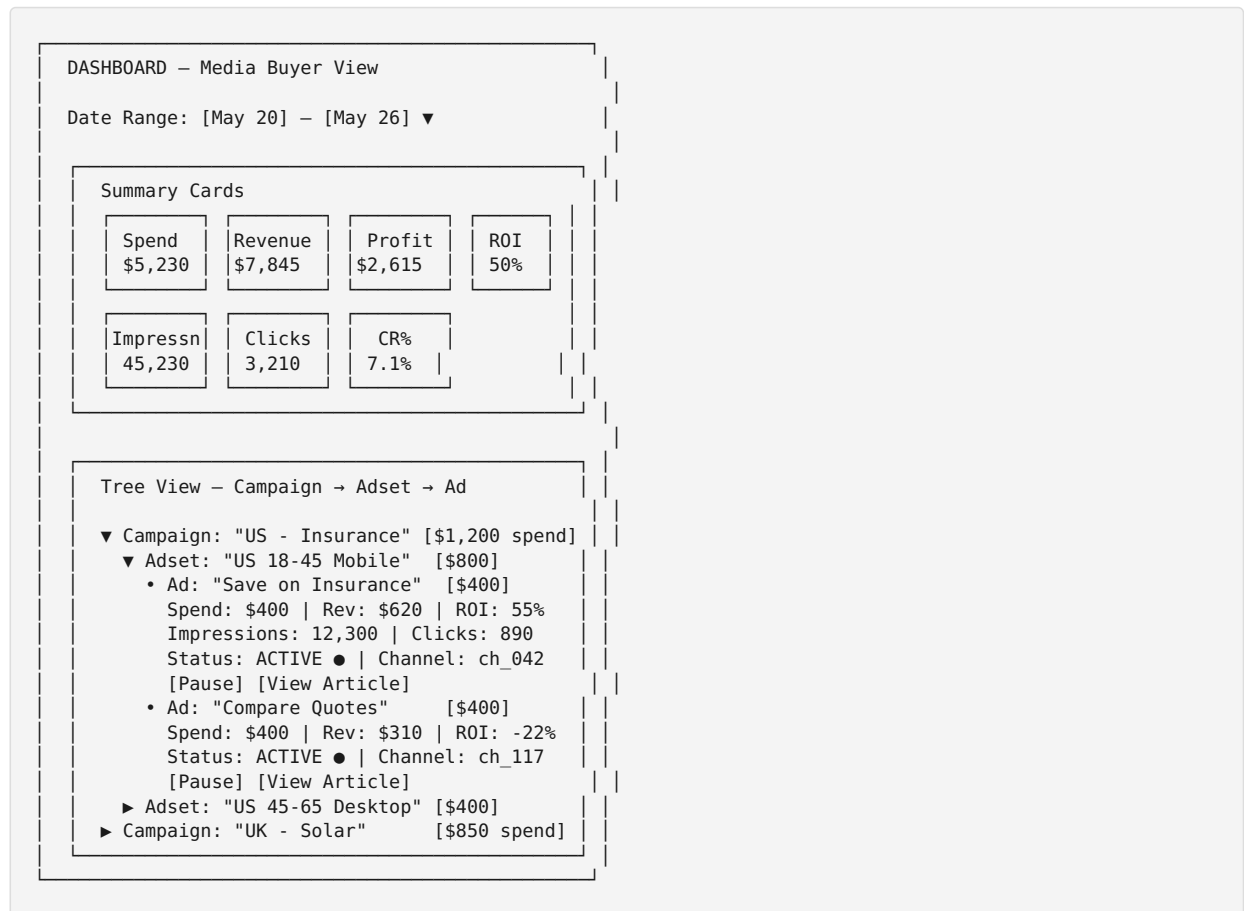
Hierarchy (highest priority first):

1. Ad-level fallback URL - set per ad in the ad launcher (same as display link)
2. Account-level fallback URL - default in user account settings

The fallback URL is also used as the Facebook ad display link, so non-ad traffic sees and lands on this URL instead of the monetized article.

5.8 Revenue Dashboard

5.8.1 Dashboard Layout



5.8.2 Metrics Available

MetricSourceUpdate Frequency

|-----|-----|-----|

ImpressionsFacebook Marketing APIHourly

ClicksFacebook Marketing APIHourly

SpendFacebook Marketing APIHourly

CPCCalculated: Spend/ClicksHourly

CPACalculated: Spend/ConversionsHourly

Views (article)Internal redirect trackingReal-time

Conversions (AFS clicks)AdSense AFS APIHourly

RevenueAdSense AFS APIHourly

AFS ClicksAdSense AFS APIHourly (suppressed if < 10/day/channel, shows "< 10")

CR%Calculated: Conversions/ViewsHourly

RPCCalculated: Revenue/ClicksHourly

RPMCalculated: Revenue/Impressions*1000Hourly

ProfitCalculated: Revenue - SpendHourly

ROICalculated: (Revenue-Spend)/Spend*100Hourly

Data Finalization:

- Facebook spend finalizes in ~48h. System re-pulls last 3 days every 6h.
- AdSense revenue finalizes in ~7 days. System re-pulls last 8 days every 6h.
- Dashboard shows disclaimer: "Revenue data finalizes in X days"
- All data stored locally in PostgreSQL. Dashboard NEVER makes live API calls.

ROI Calculated: $(\text{Revenue} - \text{Spend}) / \text{Spend} * 100$ Hourly

5.8.3 Data Aggregation Levels

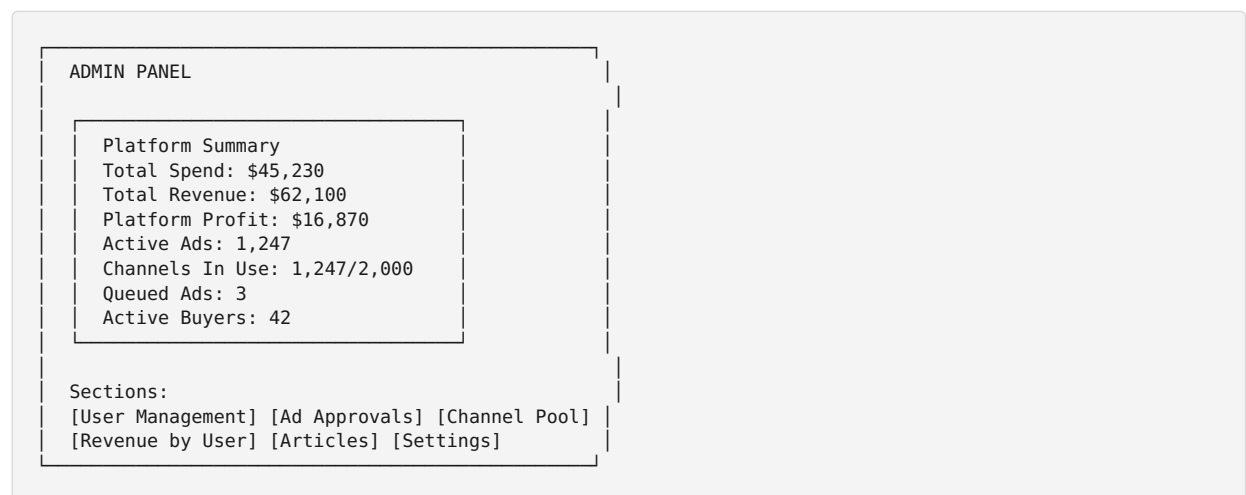
Level Grouped By
 |-----|-----|
 Campaign campaign_id
 Adset adset_id
 Ad ad_id
 Daily date
 Country geo
 Overall total

5.8.4 Buyer Actions from Dashboard

- **Pause Ad** → Calls Facebook API to pause, updates internal status
- **Resume Ad** → Calls Facebook API to resume, updates internal status
- **View Article** → Opens article page in new tab
- **Export Data** → CSV download of filtered stats

5.9 Admin Panel

5.9.1 Admin Dashboard



5.9.2 Admin Features

Feature Description

|-----|-----|

User Management Approve/reject signups, suspend users, view user list

Ad Approvals Review pending ads, approve/reject with reason

Channel Pool View all channels, status, assignment history, add/remove channels

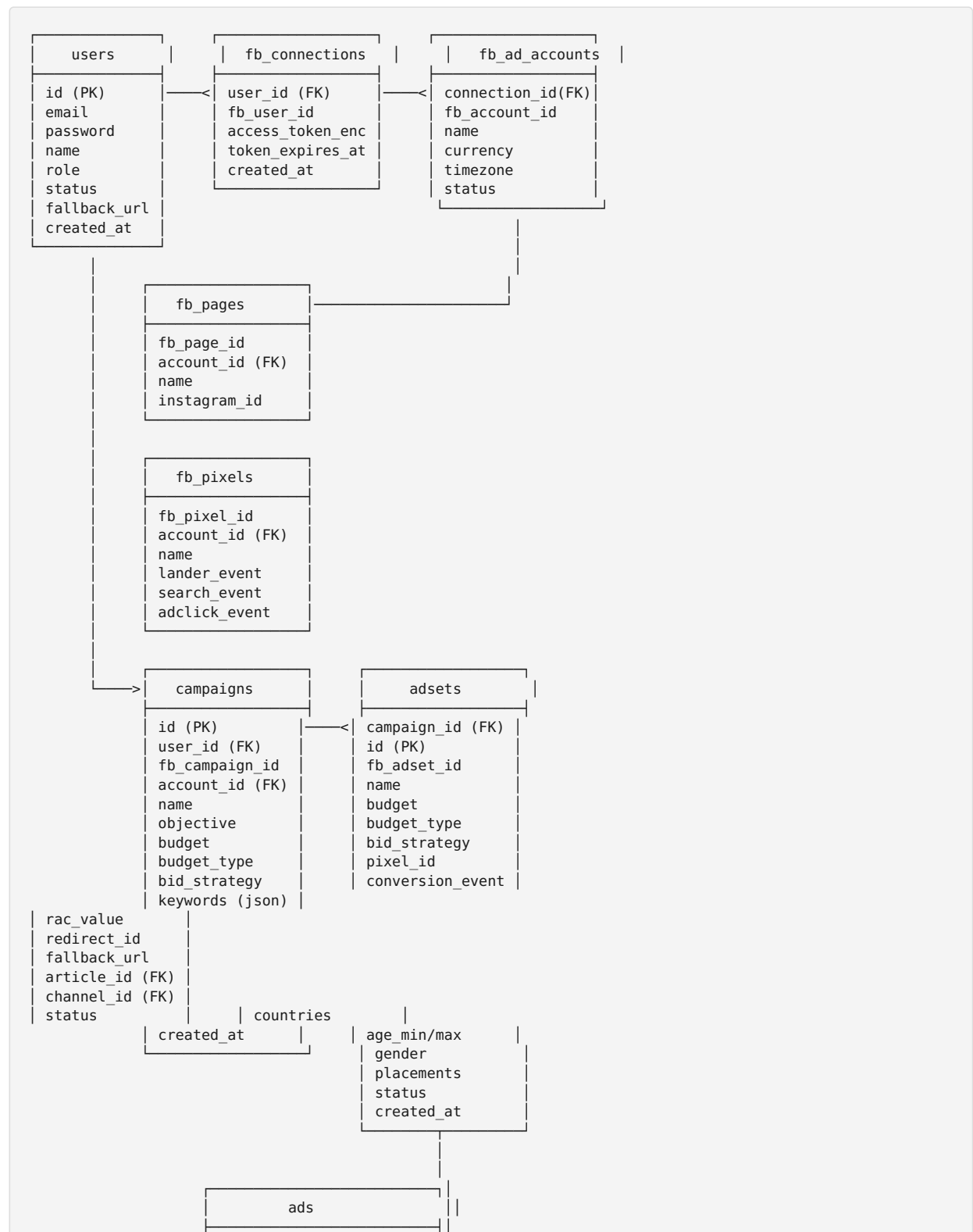
Revenue by User Table: user, total spend, total revenue, ROI, active ads

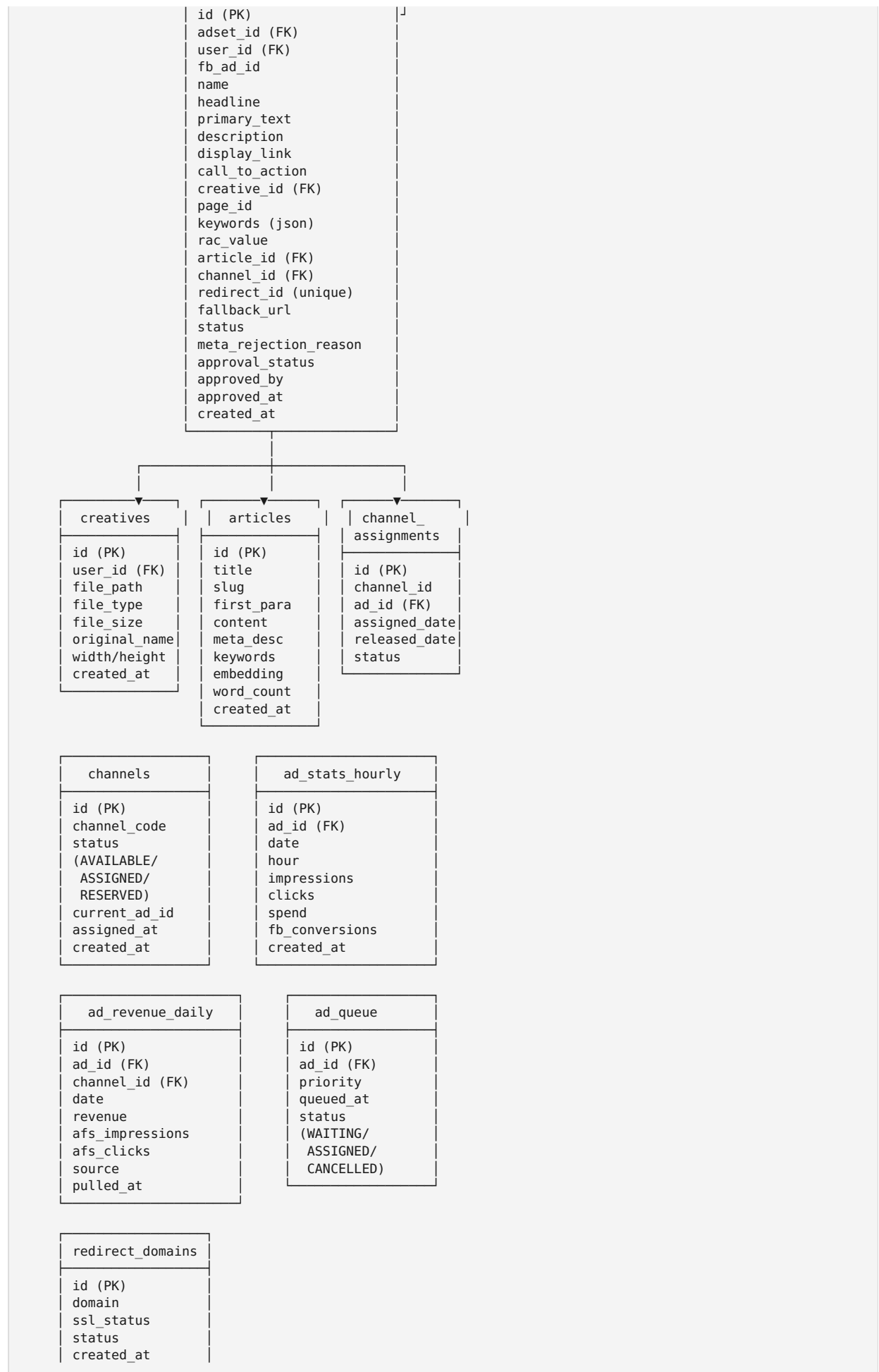
Articles View all generated articles, edit, delete

SettingsAuto-approve toggle, default fallback URL, API keys, domain config
Override AdsPause/kill any buyer's ad
Audit LogAll admin actions logged

6. Database Schema

6.1 Entity Relationship Diagram (Text)





| campaign_traffic_splits |
|-------------------------|
| id (PK) |
| campaign_id (FK) |
| website_url |
| weight_pct |
| afs_config (json) |
| status |

| data_finalization_log |
|---------------------------|
| id (PK) |
| date |
| source (facebook/adSense) |
| is_finalized |
| last_pulled_at |

| platform_settings |
|-------------------|
| key (PK) |
| value (text) |
| updated_at |
| updated_by |

| audit_log |
|----------------|
| id (PK) |
| user_id (FK) |
| action |
| entity_type |
| entity_id |
| details (json) |
| ip_address |
| created_at |

6.2 Key Indexes

```
-- Performance-critical queries
CREATE INDEX idx_ads_user_status ON ads(user_id, status);
CREATE INDEX idx_ads_channel ON ads(channel_id);
CREATE INDEX idx_ads_redirect ON ads(redirect_id);
CREATE INDEX idx_channel_assignments_date ON channel_assignments(assigned_date, channel_id);
CREATE INDEX idx_ad_stats_hourly_ad_date ON ad_stats_hourly(ad_id, date, hour);
CREATE INDEX idx_ad_revenue_daily_ad_date ON ad_revenue_daily(ad_id, date);
CREATE INDEX idx_channels_status ON channels(status);
CREATE INDEX idx_articles_embedding ON articles USING ivfflat(embedding vector_cosine_ops);
```

7. API Specification

7.1 Authentication

- POST /api/auth/signup
 POST /api/auth/login
 POST /api/auth/refresh
 POST /api/auth/logout
- Register new user
 – Login, returns JWT
 – Refresh access token
 – Invalidate refresh token

7.2 Facebook

```
GET /api/facebook/auth-url    - Get Facebook OAuth URL
GET /api/facebook/callback    - OAuth callback handler
GET /api/facebook/accounts    - List connected ad accounts
GET /api/facebook/accounts/:id/pages - List pages
GET /api/facebook/accounts/:id/pixels - List pixels
```

7.3 Campaigns

```
POST /api/campaigns          - Create campaign
GET /api/campaigns           - List user's campaigns
GET /api/campaigns/:id       - Get campaign details
PATCH /api/campaigns/:id     - Update campaign
GET /api/campaigns/:id/traffic-split - Get traffic split config
POST /api/campaigns/:id/traffic-split - Set traffic split config
```

7.4 Adsets

```
POST /api/campaigns/:id/adsets - Create adset
GET /api/campaigns/:id/adsets  - List adsets in campaign
GET /api/adsets/:id            - Get adset details
PATCH /api/adsets/:id         - Update adset
```

7.5 Ads

```
POST /api/adsets/:id/ads      - Create ad(s) in adset
GET /api/adsets/:id/ads       - List ads in adset
GET /api/ads/:id              - Get ad details
PATCH /api/ads/:id           - Update ad
POST /api/ads/:id/pause       - Pause ad
POST /api/ads/:id/resume      - Resume ad
```

7.6 Media

```
POST /api/media/upload        - Upload creative (multipart)
GET /api/media                - List user's creatives
DELETE /api/media/:id         - Delete creative
```

7.7 Stats

```
GET /api/stats                - Get aggregated stats
  ?dateStart=2026-05-20
  &dateEnd=2026-05-26
  &dimensions=campaign,adset,ad
  &metrics=spend,revenue,roi,impressions
  &groupBy=date
```

7.8 Admin

```
GET /api/admin/users          - List all users
PATCH /api/admin/users/:id    - Approve/reject/suspend user
GET /api/admin/approvals      - List pending ad approvals
POST /api/admin/approvals/:id - Approve/reject ad
GET /api/admin/channels       - List all channels
POST /api/admin/channels      - Add channels (bulk)
DELETE /api/admin/channels/:id - Remove channel
GET /api/admin/stats           - Platform-wide stats
```

```

GET /api/admin/stats/users - Per-user breakdown
POST /api/admin/ads/:id/override - Pause/kill any ad
GET /api/admin/articles - List all articles
GET /api/admin/redirect-domains - List redirect domains
POST /api/admin/redirect-domains - Add redirect domain
DEL /api/admin/redirect-domains/:id - Remove redirect domain
GET /api/admin/settings - Get platform settings
PATCH /api/admin/settings - Update settings (compliance prompt, etc.)
PATCH /api/admin/users/:id/revenue-cut - Set revenue cut % for user
PATCH /api/admin/users/:id/redirect-domain - Assign redirect domain
PATCH /api/admin/users/:id/style-id - Set custom search style ID

```

7.9 Redirect (Public — no auth)

```

GET /go/:redirect_id - Redirect engine endpoint

```

7.10 Article (Public — no auth)

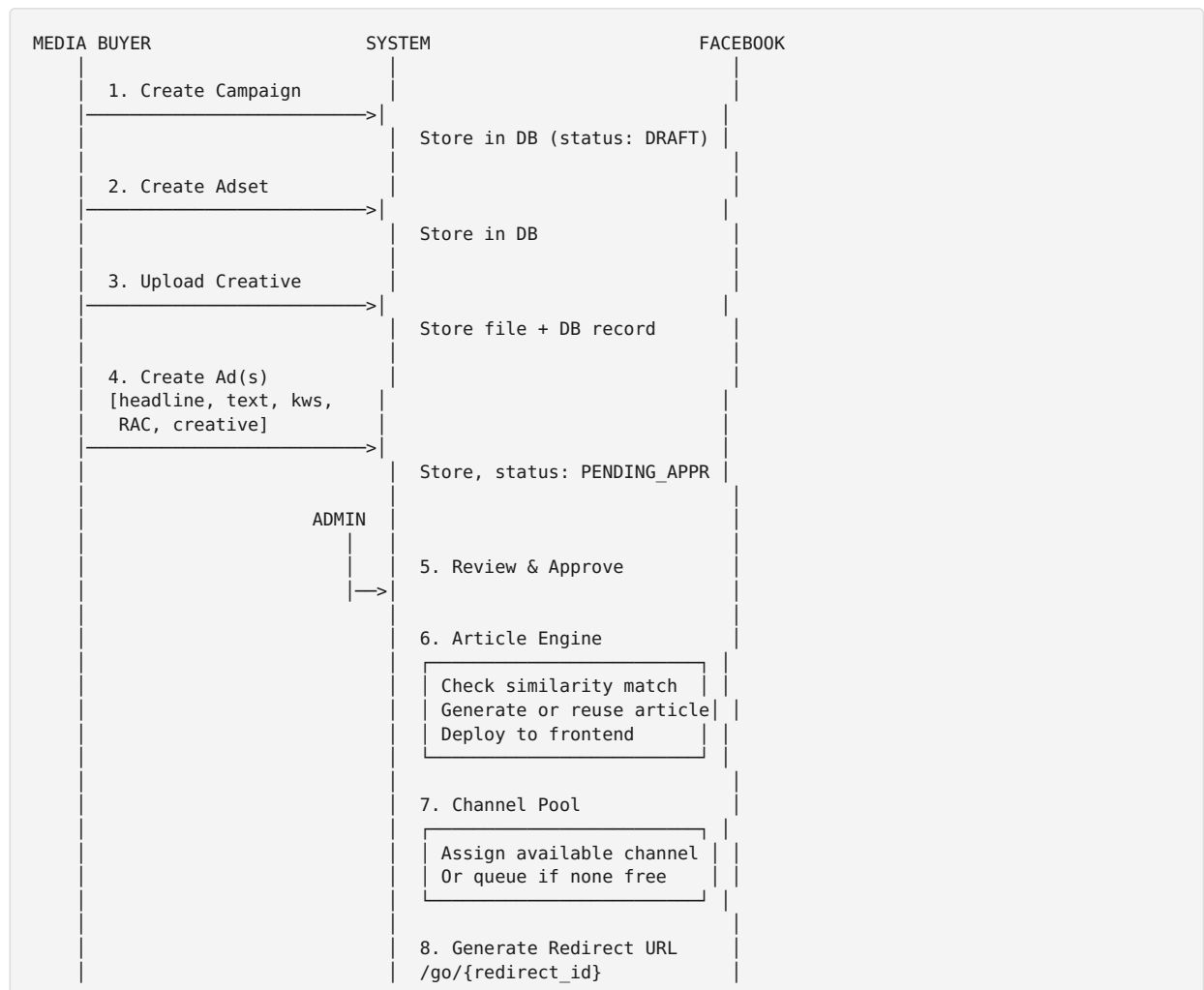
```

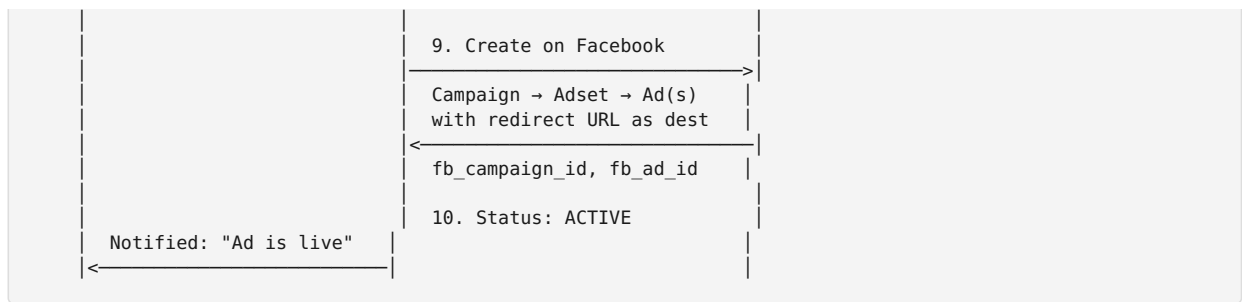
GET /a/:article_slug - Serve article page

```

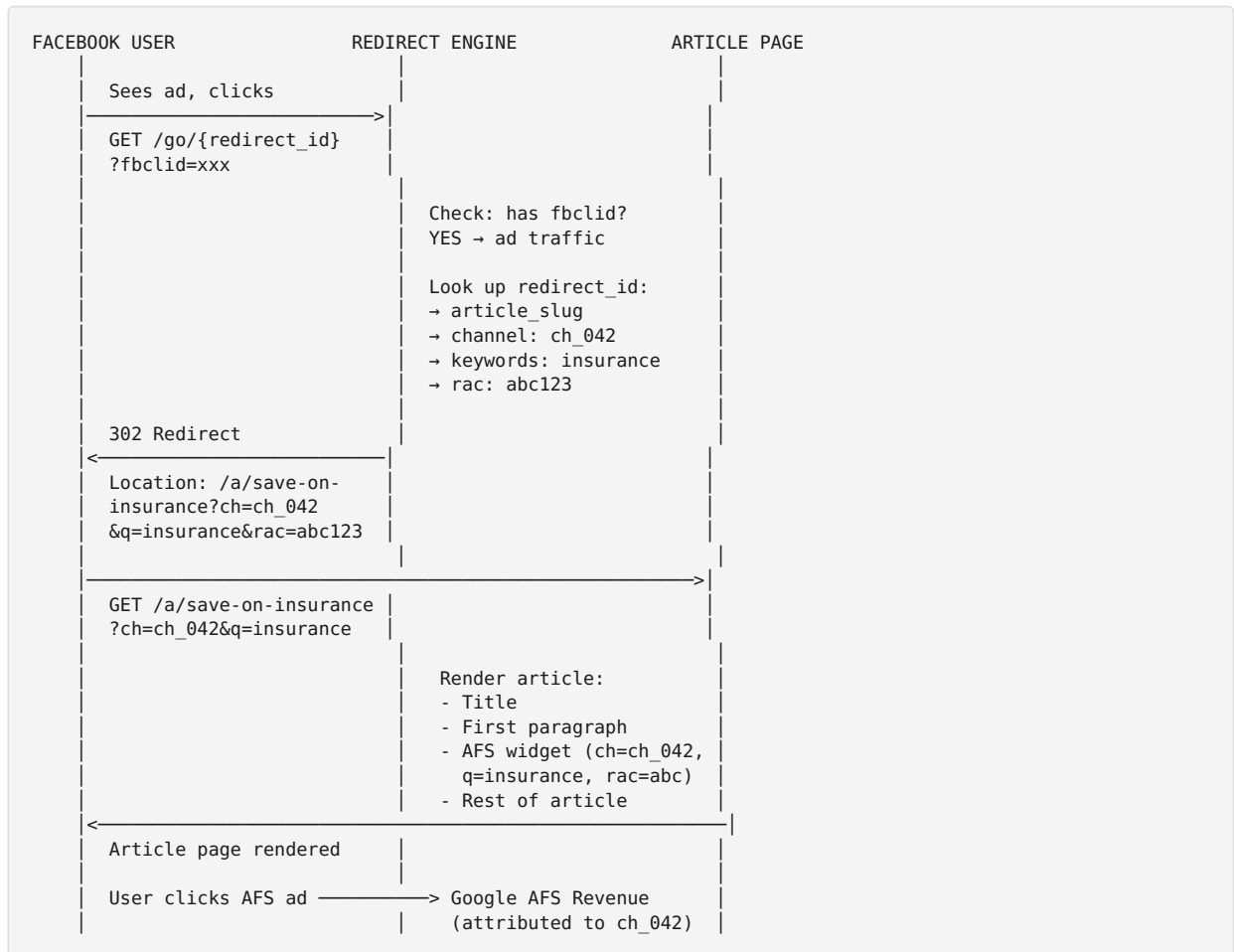
8. Flow Diagrams

8.1 Complete Ad Launch Flow

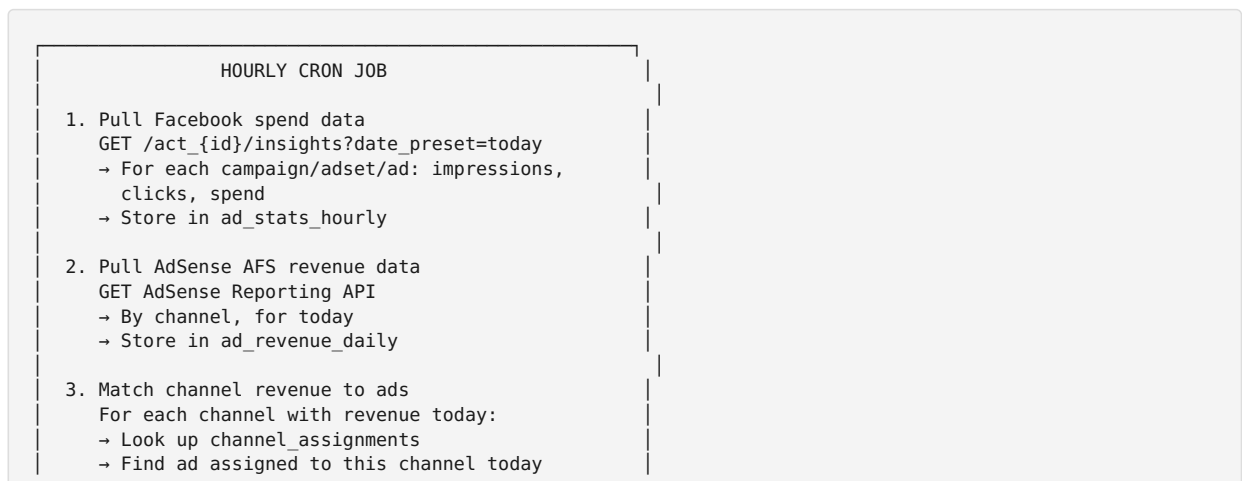




8.2 User Click Flow

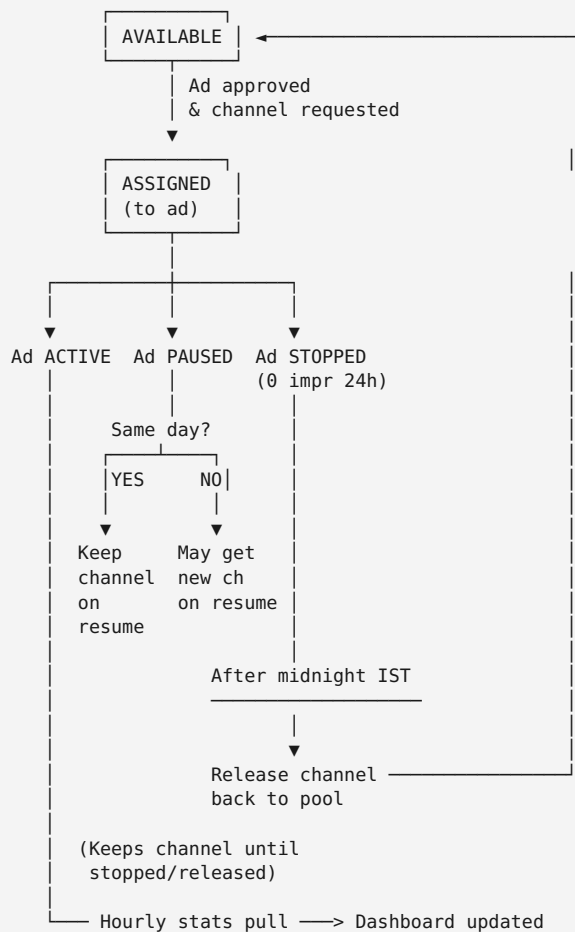


8.3 Revenue Attribution Flow



- Write to `ad_revenue_daily(ad_id, date, rev)`
- 4. Calculate derived metrics
For each ad with data today:
 - `profit = revenue - spend`
 - `roi = profit / spend * 100`
 - `cr = conversions / views * 100`
 - Store in stats cache (Redis)
- 5. Update dashboard
 - Invalidate cached stats
 - Clients auto-refresh via polling/websocket

8.4 Channel Pool Lifecycle



9. Tech Stack

LayerTechnologyRationale

|-----|-----|-----|

FrontendNext.js 14 (React)SSR for article pages, React for dashboard SPA

Backend APINode.js + ExpressJavaScript ecosystem, good for async I/O, Facebook SDK support

DatabasePostgreSQL 16Relational data, JSONB for flexible fields, pgvector for embeddings

Cache / QueueRedisSession store, stats cache, job queues (BullMQ)

Job SchedulerBullMQ + node-cronHourly stats pull, midnight cleanup, token refresh

ORMPrismaType-safe DB access, migrations

AuthJWT (jsonwebtoken)Stateless auth with refresh tokens

Facebook APIfacebook-nodejs-business-sdkOfficial Meta SDK

AdSense APIgoogleapis (Node.js)Official Google client library
AI (Articles)Claude API / OpenAI APIArticle generation
EmbeddingsOpenAI text-embedding-3-smallArticle similarity matching
File StorageLocal disk + CDN (optional)Creative uploads
Reverse ProxyNginxSSL, static files, rate limiting
Process ManagerPM2Process management, auto-restart
MonitoringPM2 metrics + custom health endpointBasic monitoring

9.5 Performance, Indexing & Caching Strategy

9.5.1 Database Indexing

Primary Query Patterns and Indexes:

```
-- User's campaigns/ads (dashboard loads)
CREATE INDEX idx_campaigns_user ON campaigns(user_id, status);
CREATE INDEX idx_adsets_campaign ON adsets(campaign_id, status);
CREATE INDEX idx_ads_adset ON ads(adset_id, status);
CREATE INDEX idx_ads_user_status ON ads(user_id, status);

-- Redirect engine (MUST be fast - under 10ms)
CREATE UNIQUE INDEX idx_ads_redirect_id ON ads(redirect_id);

-- Channel pool lookups
CREATE INDEX idx_channels_status ON channels(status);
CREATE INDEX idx_channels_current_ad ON channels(current_ad_id) WHERE current_ad_id IS NOT NULL;
CREATE INDEX idx_channel_assignments_active ON channel_assignments(channel_id, assigned_date) WHERE released_

-- Revenue attribution (hourly cron)
CREATE INDEX idx_ad_revenue_daily_ad_date ON ad_revenue_daily(ad_id, date);
CREATE INDEX idx_ad_revenue_daily_channel ON ad_revenue_daily(channel_id, date);
CREATE INDEX idx_ad_stats_hourly_composite ON ad_stats_hourly(ad_id, date, hour);

-- Stats dashboard with covering index
CREATE INDEX idx_ad_stats_user_date ON ad_stats_hourly(ad_id, date) INCLUDE (impressions, clicks, spend);

-- Article similarity search (vector index)
CREATE INDEX idx_articles_embedding ON articles USING ivfflat(embedding vector_cosine_ops) WITH (lists = 100)

-- Ad queue (FIFO)
CREATE INDEX idx_ad_queue_status ON ad_queue(status, queued_at) WHERE status = 'WAITING';
```

9.5.2 Redis Caching Strategy

Cache Key PatternDataTTLInvalidation

|---|---|---|---|

redirect:{redirect_id}ad_id, article_slug, channel, keywords, rac, fallback_url5 minOn ad update/pause
stats:user:{user_id}:{date}Aggregated daily stats for user10 minOn hourly stats pull
stats:ad:{ad_id}:{date}Per-ad daily stats10 minOn hourly stats pull
stats:admin:platform:{date}Platform-wide stats10 minOn hourly stats pull
channel:pool:available_countCount of available channels1 minOn assign/release
user:fb_accounts:{user_id}Facebook ad accounts1 hourOn reconnect
article:slug:{slug}Article content30 minOn article edit

Critical Path - Redirect Engine cached aggressively:

- Redis HIT: under 5ms response
- Redis MISS: ~20ms (PostgreSQL lookup, then cache)

9.5.3 Dashboard Stats Pre-aggregation

Instead of joining tables on every request:

1. Hourly Cron pulls Facebook stats into ad_stats_hourly (append-only)
2. Same cron pulls AdSense revenue into ad_revenue_daily

3. Derived metrics calculated and cached in Redis
4. Dashboard reads from Redis cache (10 min TTL)
5. Cache miss triggers aggregation query, result cached

Admin dashboard uses materialized view refreshed every 10 minutes.

9.5.4 Connection Pooling

- PostgreSQL: PgBouncer or Prisma pool, 20 connections (API) + 5 (worker)
- Redis: ioredis pool, 10 connections, exponential backoff reconnect

9.5.5 Rate Limiting

- API general: 100 req/min per user
- Ad creation: 20 req/min per user
- Media upload: 10 req/min per user
- Stats: 30 req/min per user
- Redirect engine: 10,000 req/min global (no per-user limit)

Implementation: Redis-based sliding window (express-rate-limit + rate-limit-redis)

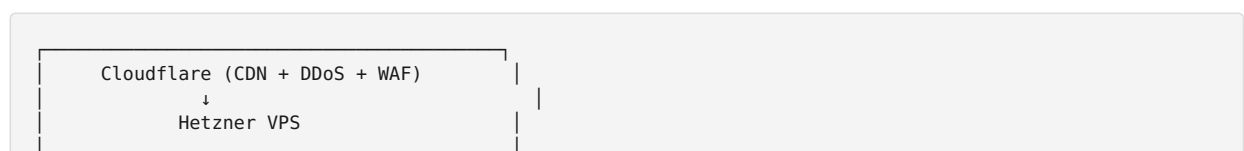
9.5.6 Scaling Path

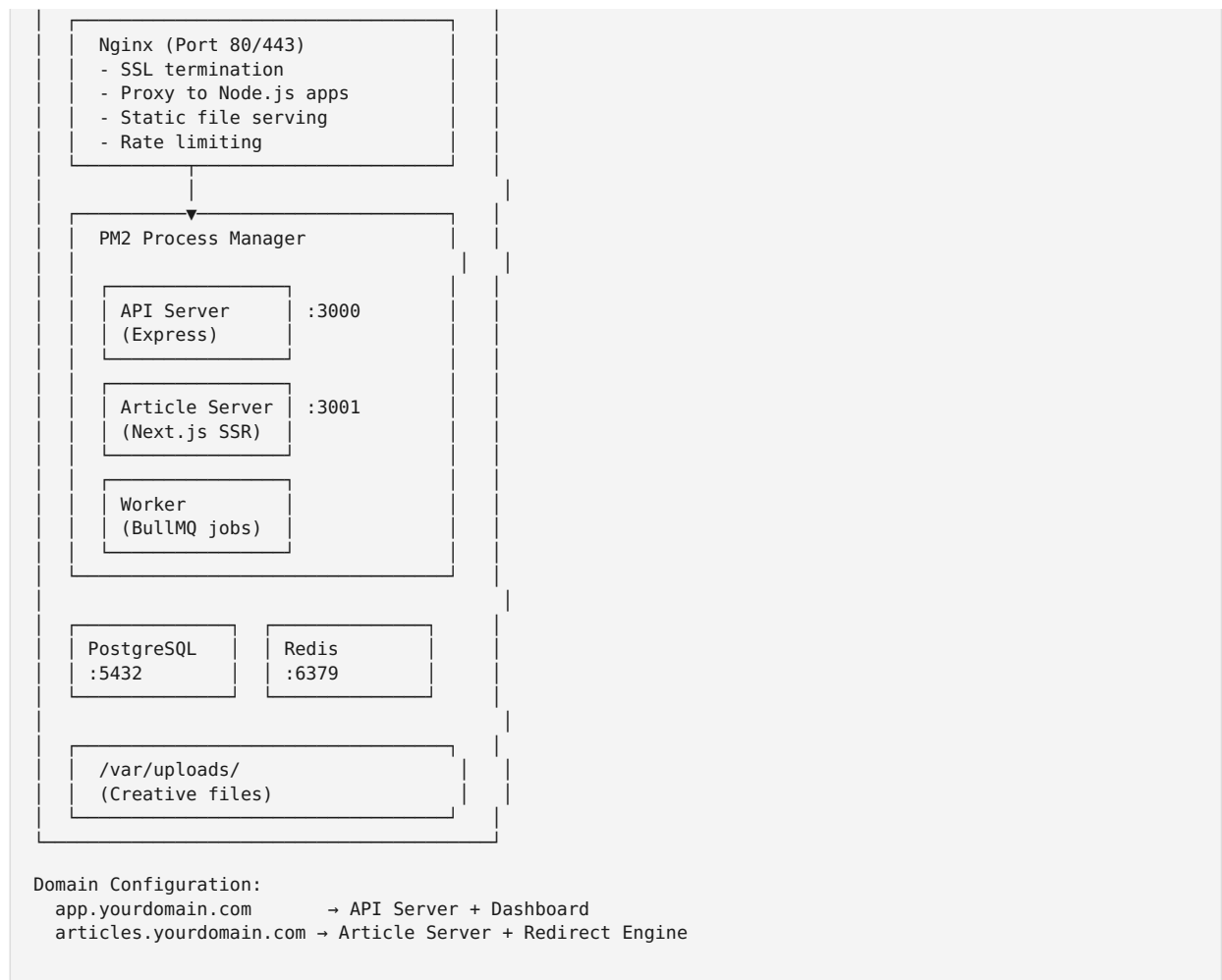
Component 50 buyers 200 buyers Action
 |---|---|---|---|
 API Server 1 instance 2-3 instances PM2 cluster mode
 Redirect Engine Same server Dedicated server Separate deploy
 PostgreSQL Single Read replicas for stats Add replica
 Redis Single Sentinel for HA Add sentinel
 Worker Single Keep single Distributed locks
 Article Frontend Same server CDN + edge cache Cloudflare

10. Non-Functional Requirements

Requirement Target
 |-----|-----|
Response Time API: < 200ms (p95), Redirect: < 50ms
Availability 99.5% uptime
Concurrent Users 50 media buyers simultaneously
Active Ads Support up to 2,000 concurrent active ads
Data Retention Stats: 12 months, Articles: indefinite
Security HTTPS only, encrypted tokens, bcrypt passwords, input sanitization
Rate Limiting API: 100 req/min per user, Redirect: 10,000 req/min
Backup Daily PostgreSQL backup, 7-day retention
GDPR User data deletion on request

11. Deployment Architecture





12. Open Questions & Future Scope

Open Questions (Need Decision)

#QuestionOptions

|---|-----|-----|

- 1Revenue margin — does platform keep a % of revenue before showing to buyer?Flat fee / Percentage / Show actual
- 2Article domain — what domain will be used?Needs to be AdSense-approved
- 3Platform domain — what domain for the dashboard?Can be subdomain of article domain
- 4Multiple article domains in future?Architecture supports it, but v1 = one domain
- 5Facebook App — need to create new app at developers.facebook.comRequires business verification

Future Scope (Not in v1)

FeaturePriority

|-----|-----|

- AI Ad Review EngineHigh — replace manual approval
- Alerting SystemMedium — ROI drops, CPA thresholds
- Bulk Ad Import (CSV)Medium
- A/B Testing FrameworkMedium — test different articles per ad
- Multi-domain Article RotationMedium
- RAC Suggestion AILow — analyze creative to suggest RAC
- Auto-optimizationLow — auto-pause/scale based on ROI
- Billing / InvoicingLow — if platform charges media buyers
- 2FA AuthenticationLow

Webhook NotificationsLow — real-time event push

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